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$$\int \frac{x}{x^4 + 2x^2 + 2} dx = \int \frac{x}{(x^2 + 1)^2} dx = \frac{1}{2} \int \frac{1}{(x^2 + 1)^2 + 1} d(x^2 + 1) \\ \stackrel{t=x^2+1}{=} \frac{1}{2} \int \frac{1}{t^2 + 1} dt = \frac{1}{2} \operatorname{Bgtan} t + c \stackrel{t=x^2+1}{=} \frac{1}{2} \operatorname{Bgtan} (x^2 + 1) + c$$

References